



Statistics of Financial Markets

Chapter 2: Classical distributions and stylised facts



Daniel Traian PELE

Bucharest University of Economic Studies

IDA Institute Digital Assets

Blockchain Research Center

AI4EFin Artificial Intelligence for Energy Finance

Romanian Academy, Institute for Economic Forecasting

MSCA Digital Finance

Acronyms used in this material

- ▣ ACF: Autocorrelation Function
- ▣ AI: Artificial Intelligence
- ▣ AIC: Akaike Information Criterion
- ▣ ARCH: AutoRegressive Conditional Heteroskedasticity
- ▣ BET: Bucharest Exchange Trading (index)
- ▣ BVB: Bursa de Valori București (the Bucharest Stock Exchange)
- ▣ CDF: Cumulative Distribution Function
- ▣ CLT: Central Limit Theorem
- ▣ EGARCH: Exponential GARCH
- ▣ ES: Expected Shortfall
- ▣ GARCH: Generalised AutoRegressive Conditional Heteroskedasticity
- ▣ GJR-GARCH: Glosten–Jagannathan–Runkle GARCH
- ▣ JB: Jarque–Bera (test)
- ▣ LB: Ljung–Box (test)
- ▣ MLE: Maximum Likelihood Estimation
- ▣ OMV: Österreichische Mineralölverwaltung (Austrian Mineral Oil Administration (OMV group))
- ▣ OUG: Ordonanță de Urgență a Guvernului (government emergency ordinance)
- ▣ QQ: Quantile–Quantile (plot)
- ▣ SFE: Statistics of Financial Markets, Exercises (Quantlet collection of the textbook)

Today's Question and Route

- **Question:** is the Normal distribution a good model for daily returns, and what do real returns look like?
 - ▶ the answer decides how we measure risk, price options and test hypotheses
- **Route** of the chapter
 - ▶ the Normal and the lognormal distributions, the central limit theorem (CLT)
 - ▶ moments, the Jarque–Bera test and QQ plots
 - ▶ the Student- t distribution as a first heavy-tailed alternative
 - ▶ six stylised facts of returns [Cont \(2001\)](#)
- Real data, 2010–2026: S&P 500, DAX, BET, Bitcoin and four BVB stocks (SNP, BRD, SNN, SNG)

Learning Outcomes

- ▣ Compute probabilities and quantiles of the Normal and lognormal distributions
- ▣ State the CLT and say when the Normal approximation works
- ▣ Estimate skewness and excess kurtosis and test normality with the Jarque–Bera test
- ▣ Read a QQ plot and fit a Student- t distribution by maximum likelihood
- ▣ Check six stylised facts on a return series and explain what each means for a model

Reading and Tools

- Textbook: [Franke, Härdle & Hafner \(2019\)](#), *Statistics of Financial Markets*, 5th ed., Sec. 3.3 (Normal and lognormal) and Sec. 11.2 (stylised facts)
 - ▶ exercises with solutions: [Borak, Härdle & López-Cabrera \(2013\)](#)
- Complementary: [Tsay \(2010\)](#), Sec. 1.2 (distributional properties of returns); [Cont \(2001\)](#)
- Python Quantlets of this chapter: [Quantlets/Ch_02](#)
 - ▶ ported from the SFE Quantlets of the textbook (SFEDaxReturnDistribution, SFElognormal, SFEclt, SFENormalApprox1-4)
 - ▶ each chart in these slides links to the Quantlet that draws it
- Lecture notebook: [open in Google Colab](#)
- Video course:  [Tukey's g- and h- transformations](#)

A Day That Should Not Happen

- 16 March 2020: the S&P 500 fell by -12.8% (log return) in one day
- Daily standard deviation 2010–2026: 1.08% , so the fall was -11.8 standard deviations
 - ▶ under the Normal distribution, the probability of such a day is about 2×10^{-32}
 - ▶ that is, less than once in 10^{29} years of trading
- Yet such days happen: 19 October 1987, October 2008, March 2020, April 2025
- Conclusion: the model, not the market, is wrong

Question for the Room: How Often Is a 4-Sigma Day?

- ▣ A **k -sigma day**: a daily return more than k standard deviations away from the mean
- ▣ **What do you think?** If daily returns were Normal, how often would a 4-sigma day occur?
- ▣ **What do you think?** How often does it occur in the S&P 500 since 2010?
- ▣ **Answer**
 - ▶ Normal distribution: once every 15 787 trading days, about once in 63 years
 - ▶ S&P 500: 28 days in 4 201 days, once every 150 days, about 1.7 per year

What a Distribution Is Used For

- ▣ **Risk:** the probability of a large loss
 - ▶ VaR 1% (Value at Risk): the loss exceeded with probability 1%; a quantile of the distribution ([Chapter 10](#))
- ▣ **Pricing:** the Black–Scholes formula assumes lognormal prices
 - ▶ heavier tails make deep out-of-the-money options more valuable
- ▣ **Inference:** t -tests and confidence intervals assume (approximate) normality
 - ▶ heavy tails make standard errors unreliable in small samples
- ▣ A wrong distribution gives wrong numbers, even with perfect data

The Benchmark Model of This Chapter

- Price P_t , log return $r_t = \ln(P_t/P_{t-1})$, so $P_t = P_{t-1}e^{r_t} = P_0e^{r_1+\dots+r_t}$
 - ▶ notation: **i.i.d.** = independent and identically distributed
- **Benchmark:** r_t i.i.d. $N(\mu, \sigma^2)$
 - ▶ prices follow a geometric random walk; P_t is lognormal [Osborne \(1959\)](#)
 - ▶ the first model of this kind, for arithmetic price changes: [Bachelier \(1900\)](#)
- This chapter tests each assumption of the benchmark in turn: normality, identical distribution and independence
- Textbook: [Franke, Härdle & Hafner \(2019\)](#), Sec. 3.3 and Sec. 11.2

Carl Friedrich Gauss and the Normal Distribution

- ▣ De Moivre (1733) found the bell curve as the limit of binomial probabilities
- ▣ Gauss (1809) used it for the errors of astronomical observations and derived least squares from it
- ▣ Laplace (1810) proved a first version of the central limit theorem
- ▣ Hence its other name: the **Gaussian** distribution
- ▣ In finance it became the default model of returns, because sums of Normal variables stay Normal



Carl Friedrich Gauss (1777–1855)

Portrait: [Christian Albrecht Jensen \(1840\)](#); public domain; [Wikimedia Commons](#)

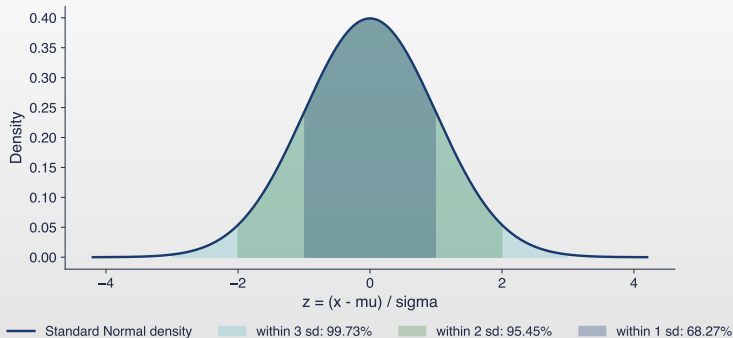
Definition

- $X \sim N(\mu, \sigma^2)$ has the **PDF** (probability density function)
 - ▶ $f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right), \quad x \in \mathbb{R}$
 - ▶ two parameters: the mean μ (location) and the variance σ^2 (spread)
- **Standard Normal distribution:** $Z = (X - \mu)/\sigma \sim N(0, 1)$
 - ▶ its **CDF** (cumulative distribution function) is $\Phi(z) = P(Z \leq z)$, with no closed form; tables or software
- Probabilities of X come from Z : $P(X \leq x) = \Phi((x - \mu)/\sigma)$
- Quantile of order α : $x_\alpha = \mu + \sigma z_\alpha$, with $z_{0.01} = -2.33$, $z_{0.05} = -1.64$

Properties Used in Finance

- Symmetric around μ : mean = median = mode
- Skewness 0 and kurtosis 3 (defined in Section 5)
- **Closed under addition:** if $X \sim N(\mu_1, \sigma_1^2)$ and $Y \sim N(\mu_2, \sigma_2^2)$ are independent, then $X + Y \sim N(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2)$
 - ▶ h -day log return under the benchmark: $r_t(h) \sim N(h\mu, h\sigma^2)$
 - ▶ volatility grows with $\sqrt{h}$: the square-root-of-time rule used in [Chapter 1](#)
- Linear combinations of jointly Normal returns are Normal, so portfolio returns stay Normal
- Fully described by two numbers: once μ and σ are known, every probability is known

The 68–95–99.7 Rule



- Within 1, 2 and 3 standard deviations: 68.27%, 95.45% and 99.73% of the probability
- Outside 3 standard deviations: one day in 370; the tails of the Normal distribution fall off like $e^{-z^2/2}$, very fast

How Rare Are Large Moves under the Normal Distribution?

k	$P(Z > k)$	once every ... trading days	once every ... years
1	0.3173	3	0.013
2	0.0455	22	0.087
3	2.7×10^{-3}	370	1.5
4	6.3×10^{-5}	15 787	63
5	5.7×10^{-7}	1 744 278	6 922
6	2.0×10^{-9}	506 797 346	2 011 101

- ▣ Years computed with 252 trading days a year
- ▣ Each extra standard deviation makes a move 7 to 291 times rarer
- ▣ A 6-sigma day should not happen in the whole history of stock markets

Worked Example: Large Losses of the S&P 500

- S&P 500, 2010–2026: $n = 4\,201$ daily log returns, mean 0.045%, standard deviation 1.08%
- A loss larger than 3%: $z = (-3 - 0.045)/1.08 = -2.81$
 - ▶ Normal distribution: $\Phi(-2.81) = 0.25\%$, so 10.5 days expected
 - ▶ observed: 49 days
- A loss larger than 5%: $z = -4.65$
 - ▶ Normal distribution: 6.9×10^{-3} days expected; observed: 8 days
- BET: 13 days below -5% against 5.1×10^{-3} expected
- The Normal distribution underestimates the frequency of large losses by orders of magnitude

Recap: The Normal Distribution

- Two parameters, symmetric, closed under addition, volatility scales with $\sqrt{h}$
- Its tails are thin: a 4-sigma day once every 63 years
- Real returns have many more large moves than it allows

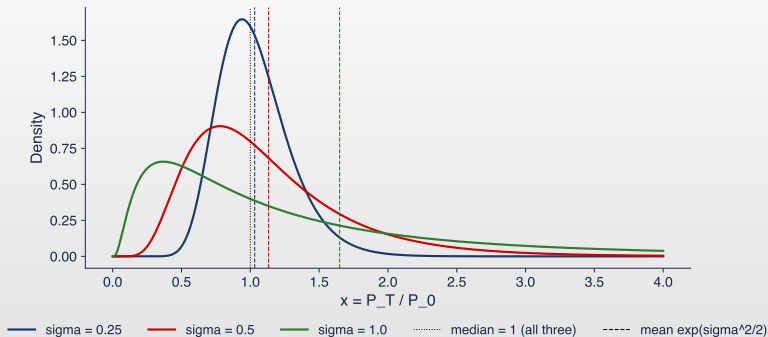
From Normal Log Returns to Lognormal Prices

- A positive variable Y is **lognormal**, $Y \sim LN(m, s^2)$, if $\ln Y \sim N(m, s^2)$
- Density: $f(y) = \frac{1}{ys\sqrt{2\pi}} \exp\left(-\frac{(\ln y - m)^2}{2s^2}\right)$, $y > 0$
- Under the benchmark, $\ln(P_T/P_0) = r_1 + \dots + r_T \sim N(T\mu, T\sigma^2)$
 - ▶ so $P_T/P_0 \sim LN(T\mu, T\sigma^2)$ [Osborne \(1959\)](#)
- Why lognormal and not Normal prices?
 - ▶ prices cannot be negative; a Normal price could be
 - ▶ the simple return $R = e^r - 1 > -1$: you cannot lose more than you invested
- Textbook: [Franke, Härdle & Hafner \(2019\)](#), Sec. 3.3; the Black–Scholes model is built on it

Mean, Median and Mode

- If $Y \sim LN(m, s^2)$:
- median e^m ; mode e^{m-s^2} ; mean $E[Y] = e^{m+s^2/2}$
 - ▶ always mode $<$ median $<$ mean: a long right tail (positive skewness)
- Variance: $\text{Var}(Y) = (e^{s^2} - 1) e^{2m+s^2}$
- The mean comes from $E[e^X] = e^{m+s^2/2}$ for $X \sim N(m, s^2)$ (Seminar 2 derives it)
 - ▶ the extra $s^2/2$ is the volatility drag of [Chapter 1](#), seen from the other side
- Which one matters? The median is the outcome of a typical investor; the mean is pulled up by a few very good paths

Lognormal Densities (SFElognormal)



- $m = 0$: the median is 1 for all three curves; the dashed lines mark the means
- $s = 0.25$: mean 1.03, mode 0.94, almost symmetric; $s = 1$: mean 1.65, mode 0.37, skewness 6.18
- Long horizons or high volatility: the gap between mean and median widens

Worked Example: 100 Lei in the BET

- BET, 2010–2026: annual mean log return $\mu = 11.6\%$ and volatility $\sigma = 16.9\%$ (daily moments $\times 251$ and $\sqrt{251}$)
- After 1 year, $P_1/P_0 \sim LN(\mu, \sigma^2)$
 - ▶ median $100e^\mu = 112.3$; mean $100e^{\mu+\sigma^2/2} = 113.9$
 - ▶ probability of a loss $\Phi(-\mu/\sigma) = 24.8\%$; 5% quantile $100e^{\mu-1.645\sigma} = 84.9$
- After 10 years, $P_{10}/P_0 \sim LN(10\mu, 10\sigma^2)$
 - ▶ median 317.7; mean 366.8; probability of a loss 1.5%; 5% quantile 131.6
- The probability of a loss falls with the horizon; after 10 years even the 5% quantile is above 100, but only if the historical mean persists
- Caveat: these numbers inherit every weakness of the Normal model of log returns

Recap: The Lognormal Distribution

- Normal log returns $\Rightarrow$ lognormal prices: positive, right-skewed
- Median e^m below mean $e^{m+s^2/2}$: the gap is the volatility drag
- Report the median and a low quantile, not only the mean

The Galton Board

- Francis Galton (1889): balls fall through rows of pins and bounce left or right with probability $1/2$
- The final position of a ball is a **sum** of many small independent shocks
- The heaps form a bell curve: binomial probabilities close to the Normal density
- The same argument was used for returns: a daily return is the sum of many small price changes during the day



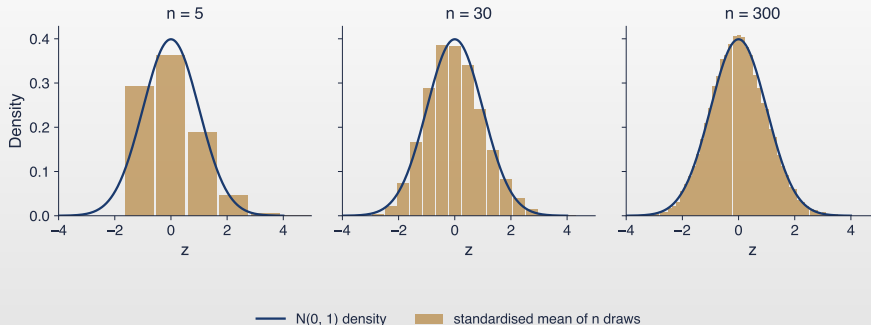
A Galton board after the balls have fallen

Photo: Klaus-Dieter Keller (2010); public domain; [Wikimedia Commons](#)

The Central Limit Theorem

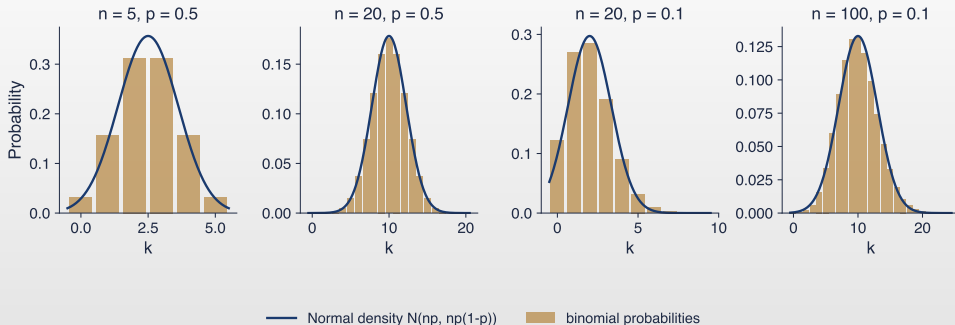
- ▣ **CLT** (Lindeberg–Lévy): $X_1, \dots, X_n$ i.i.d. with mean μ and finite variance σ^2
 - ▶ $\frac{\sqrt{n}(\bar{X}_n - \mu)}{\sigma} \xrightarrow{d} N(0, 1)$ as $n \rightarrow \infty$
 - ▶ equivalently, the sum $X_1 + \dots + X_n$ is approximately $N(n\mu, n\sigma^2)$
- ▣ Three conditions
 - ▶ independence; identical distribution; finite variance
- ▣ How fast? The error falls like $1/\sqrt{n}$ and grows with the skewness of X (Berry–Esseen bound)
- ▣ It explains why the Normal distribution is everywhere, and why sample means have Normal confidence intervals

The CLT at Work (SFEclt)



- Standardised mean of n Bernoulli(0.2) draws (a skewed variable), 20 000 simulations
- Share above 1.96: 5.8% for $n = 5$, 2.7% for $n = 30$, 2.8% for $n = 300$; Normal distribution: 2.5%
- Skewness of the mean, $(1 - 2p)/\sqrt{np(1 - p)}$: 0.67, 0.27, 0.09

The Normal Approximation of the Binomial (SFENormalApprox)



- ▣ $B(n, p) \approx N(np, np(1-p))$; with continuity correction, $P(X \leq k) \approx \Phi((k + 0.5 - np)/\sqrt{np(1-p)})$
- ▣ Largest error of the CDF: 0.001 for $n = 20, p = 0.5$; 0.037 for $n = 20, p = 0.1$; 0.017 for $n = 100, p = 0.1$
- ▣ Rule of thumb: the approximation is good when $np(1-p)$ is large (at least about 9)

Question for the Room: Does the CLT Make Daily Returns Normal?

- A daily log return is the sum of thousands of intraday log returns
- **What do you think?** Should daily returns then be close to the Normal distribution?
- **Answer:** not necessarily; each condition of the CLT can fail
 - ▶ dependence: calm and turbulent days come in clusters (Section 8)
 - ▶ changing distribution: volatility differs from day to day
 - ▶ infinite variance: Mandelbrot (1963) proposed it for cotton prices (Chapter 3, α -stable distributions)
- The data decide: the next sections measure how far returns are from the Normal distribution

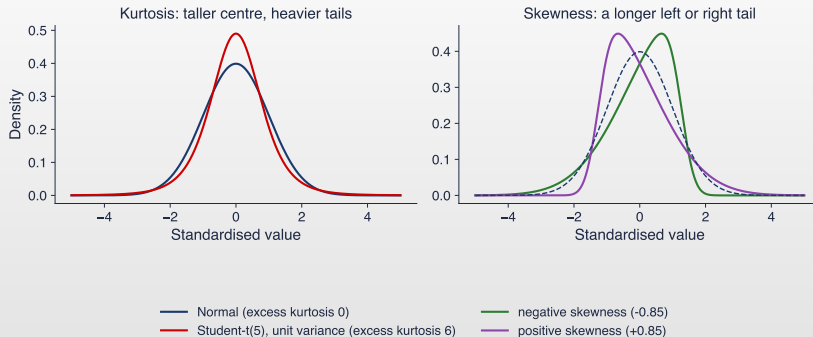
Recap: The Central Limit Theorem

- ▣ Sums of many i.i.d. terms with finite variance are approximately Normal
- ▣ Convergence is slower for skewed variables; $np(1 - p) \geq 9$ for the binomial
- ▣ Returns may violate independence, identical distribution or finite variance

Four Moments

- ▣ Mean $\mu = E[X]$; variance $\sigma^2 = E[(X - \mu)^2]$
- ▣ **Skewness:** $\gamma_1 = E[(X - \mu)^3]/\sigma^3$
 - ▶ $\gamma_1 < 0$: a longer left tail, large losses more frequent than large gains
- ▣ **Kurtosis:** $\gamma_2 = E[(X - \mu)^4]/\sigma^4$; Normal distribution: $\gamma_2 = 3$
 - ▶ **excess kurtosis** $\gamma_2 - 3$: positive means heavy tails and a tall, narrow centre (**leptokurtic**)
- ▣ Both are free of units: they describe the shape, not the location or the scale
- ▣ Fourth powers make the kurtosis very sensitive to a few extreme days

What Skewness and Kurtosis Look Like



- Left: same mean and variance; the Student- $t(5)$ curve (Section 7) has excess kurtosis 6: more mass in the centre and in the tails, less in the shoulders
- Right: skew-Normal densities with skewness ± 0.85 ; the dashed curve is the Normal distribution

Sample Moments

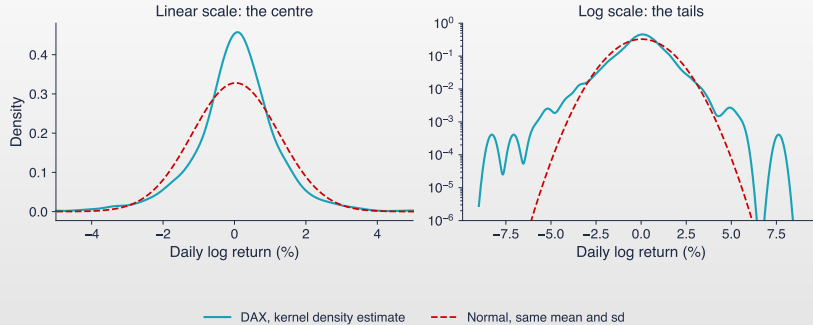
- ▣ Central sample moments: $m_k = \frac{1}{n} \sum_{t=1}^n (r_t - \bar{r})^k$
- ▣ Sample skewness $S = m_3/m_2^{3/2}$; sample excess kurtosis $K = m_4/m_2^2 - 3$
- ▣ If the data are i.i.d. Normal: $S \approx N(0, 6/n)$ and $K \approx N(0, 24/n)$
 - ▶ with $n = 4\,201$: standard errors $\sqrt{6/n} = 0.038$ and $\sqrt{24/n} = 0.076$
- ▣ These standard errors hold only under normality; for heavy-tailed data they are far too small (Seminar 2, B1)
- ▣ Textbook exercises: Borak, Härdle & López-Cabrera (2013)

Daily Log Returns: Moments, 2010–2026

Series	from	<i>n</i>	Mean %	S.d. %	Skew.	Exc. kurt.	Min %
S&P 500	2010	4 201	0.045	1.08	−0.61	13.5	−12.8
DAX	2010	4 244	0.034	1.22	−0.45	7.4	−13.1
BET	2010	4 186	0.046	1.07	−0.98	17.9	−11.9
Bitcoin	2014	4 384	0.118	3.49	−0.70	12.0	−46.5
OMV Petrom (SNP)	2010	3 815	0.076	1.62	−0.54	8.9	−15.8
BRD	2010	3 771	0.044	1.66	−0.38	7.3	−18.2
Nuclearelectrica (SNN)	2013	2 911	0.099	1.52	0.27	5.1	−9.4
Romgaz (SNG)	2013	2 909	0.090	1.40	−0.34	5.3	−11.0

- Every series has excess kurtosis between 5.1 (SNN) and 17.9 (BET): far from 0
- Every series except SNN has negative skewness
- Worst days: S&P 500 on 16 March 2020, BET on 19 December 2018 (OUG 114, a tax on bank assets), Bitcoin on 12 March 2020

DAX Returns vs the Normal Density (SFEDaxReturnDistribution)



- Kernel density estimate (a smoothed histogram) against the Normal density with the same mean and standard deviation
- Centre: peak 0.46 vs 0.33; 76.8% of days within one standard deviation, vs 68.3% for the Normal distribution
- Tails (log scale): at -6% the estimated density is 439 times the Normal one

Check the Data Before Measuring Tails

- Banca Transilvania (TLV), 27–31 May 2016: the close falls from 2.83 to 2.09 lei, the ex-date of a bonus share issue
- The adjusted close in our data: 3.60, then 3.10, then 3.81
 - ▶ log returns -15.0% and $+20.5\%$ on two consecutive days: the adjustment was applied one day late
- Excess kurtosis of TLV, 2010–2026: 17.5 with these two days, 13.5 without them
- Two wrong days out of thousands change the kurtosis by a third: this is why TLV is not in the tables of this chapter
- Rule from [Chapter 1](#): list the largest moves and check each one, especially pairs of opposite jumps

Recap: Moments

- ▣ Skewness measures asymmetry, excess kurtosis measures tail weight relative to the Normal distribution
- ▣ Daily returns: mostly negative skewness and excess kurtosis between 5 and 18
- ▣ Kurtosis is driven by a few days: check them before trusting it

The Jarque–Bera Test

- ▣ H_0 : the data are Normal, so skewness = 0 and excess kurtosis = 0
- ▣ **JB** statistic Jarque & Bera (1980), Jarque & Bera (1987): $JB = \frac{n}{6} \left(S^2 + \frac{K^2}{4} \right)$
 - ▶ the sum of two squared standardised statistics: $(S/\sqrt{6/n})^2 + (K/\sqrt{24/n})^2$
- ▣ Under H_0 and for large n : $JB \sim \chi^2(2)$; reject at 5% if $JB > 5.99$
- ▣ Simple, uses only two moments, and is the standard test reported in empirical finance

Worked Example: Jarque–Bera for the S&P 500

- ▣ $n = 4\,201$, $S = -0.61$, $K = 13.5$
- ▣ $JB = 700.2 \times (0.367 + 45.5) = 32\,113$
- ▣ Critical value 5.99: normality is rejected; the p -value is zero to any number of decimals
- ▣ The kurtosis term dominates: almost all of JB comes from $K^2/4$
- ▣ JB for the other series: BET 56 762, DAX 9 909, Bitcoin 26 482, SNN 3 174
 - ▶ all rejected at any usual level

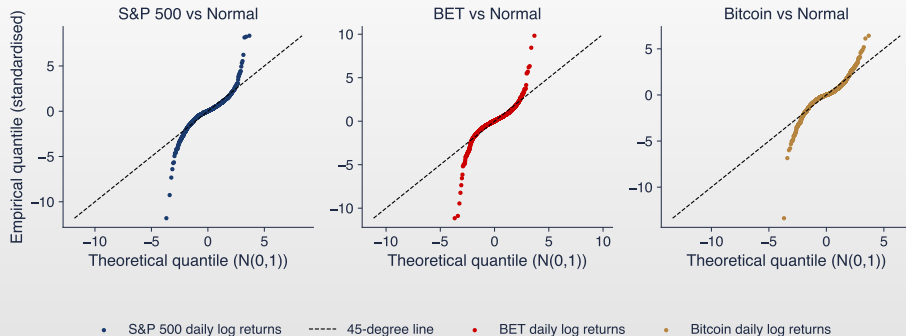
What Jarque–Bera Does Not Tell You

- ▣ With thousands of observations it rejects even tiny departures
 - ▶ report the size of S and K , not only the p -value
- ▣ It assumes i.i.d. data; with volatility clustering its standard errors are too small
 - ▶ S&P 500: Normal-theory 95% interval for K : $[13.34, 13.64]$; bootstrap interval: $[6.7, 20.6]$ (Seminar 2, B1)
- ▣ A rejection says the data are not Normal; it does not say which distribution they follow
- ▣ Other tests (Kolmogorov–Smirnov, Anderson–Darling) compare the whole CDF: [Chapter 6](#)

QQ Plots

- A **QQ plot** (quantile–quantile plot) compares the empirical quantiles with those of a model [Wilk & Gnanadesikan \(1968\)](#)
 - ▶ sort the data: $r_{(1)} \leq r_{(2)} \leq \dots \leq r_{(n)}$
 - ▶ plot $r_{(i)}$ against the model quantile $F^{-1}((i - 0.5)/n)$
- How to read it
 - ▶ points on the 45-degree line: the model fits
 - ▶ an S-shape, low end below and high end above the line: heavier tails than the model
 - ▶ only one end bent away: skewness
- Unlike JB, it shows **where** the model fails: centre, shoulders or tails

QQ Plots against the Normal Distribution



- Standardised returns against $N(0, 1)$ quantiles: all three bend away from the line at both ends
- S&P 500: lowest standardised return -11.8 where the Normal quantile is -3.7 ; BET: -11.2 vs -3.7
- The lower tail bends more than the upper one: negative skewness

Recap: Testing Normality

- ▣ JB combines skewness and kurtosis; it rejects normality for every series
- ▣ Report the size of the departure; Normal-theory standard errors are too small
- ▣ QQ plots show where the model fails: here, in both tails

"Student": William Sealy Gosset

- ▣ Chemist and brewer at Guinness, Dublin, working with small samples of barley and hops
- ▣ The brewery did not allow its staff to publish under their own names: he signed "Student" [Student \(1908\)](#)
- ▣ He derived the distribution of $(\bar{X} - \mu)/(s/\sqrt{n})$ for Normal data: the t distribution
- ▣ In finance it is used for a different reason: its tails are heavier than those of the Normal distribution [Praetz \(1972\)](#), [Blattberg & Gonedes \(1974\)](#)



William Sealy Gosset (1876–1937)

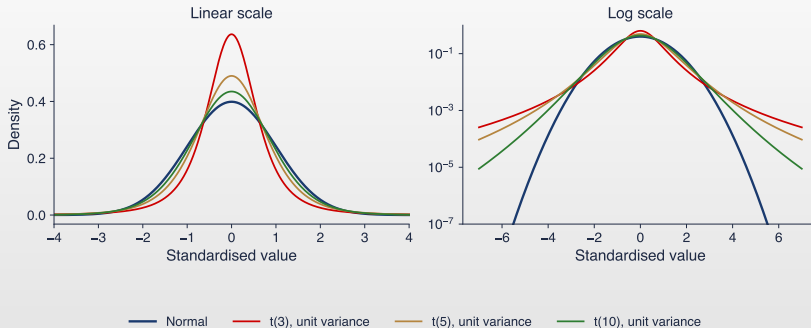
Photo: [unknown author \(1908\)](#); public domain;
[Wikimedia Commons](#)

Definition

- $T = Z/\sqrt{V/\nu}$ with $Z \sim N(0, 1)$ independent of $V \sim \chi^2(\nu)$: $T \sim t(\nu)$
 - ▶ $\nu > 0$: the **degrees of freedom**, which control the tails
- Density: $f(x) = \frac{\Gamma((\nu+1)/2)}{\sqrt{\nu\pi} \Gamma(\nu/2)} \left(1 + \frac{x^2}{\nu}\right)^{-(\nu+1)/2}$
- Location-scale version for returns: $r = m + s T$, with location m , scale s and ν
- As $\nu \rightarrow \infty$, $t(\nu) \rightarrow N(0, 1)$: the Normal distribution is a limiting case

Moments and Tails

- Mean 0 if $\nu > 1$; variance $\nu/(\nu - 2)$ if $\nu > 2$; excess kurtosis $6/(\nu - 4)$ if $\nu > 4$
- Moments of order ν and higher are infinite
 - ▶ $\nu \leq 4$: infinite kurtosis; $\nu \leq 2$: infinite variance
- **Power-law tails:** $P(|T| > x) \approx c x^{-\nu}$ for large x
 - ▶ the Normal tail falls like $e^{-x^2/2}$, much faster than any power
 - ▶ ν is also the **tail index**, estimated directly in [Chapter 5](#)
- To compare with the Normal distribution at the same variance, use $T\sqrt{(\nu - 2)/\nu}$, which has variance 1

Student- t Densities with Unit Variance

- Smaller ν : a taller centre and heavier tails; on the log scale the Normal tails drop like a parabola
- $P(|X| > 4)$ at unit variance: $t(3)$ 0.62%, $t(5)$ 0.36%, $t(10)$ 0.12%, Normal distribution 0.0063%
- With $\nu = 5$, a 4-sigma day is 56 times more likely than under the Normal distribution

Fitting a Student- t by Maximum Likelihood

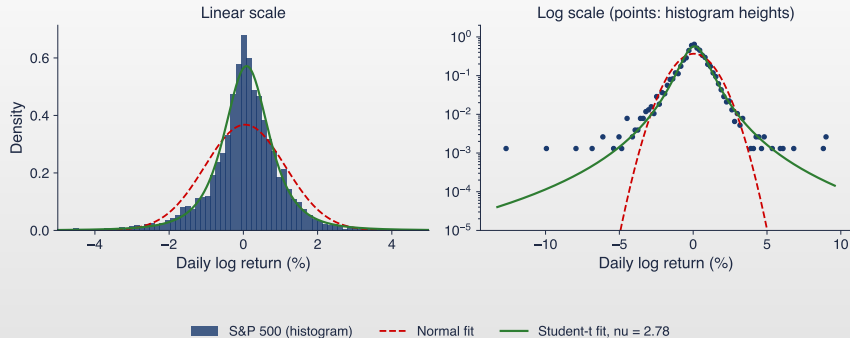
- **MLE** (maximum likelihood estimation): choose (ν, m, s) that maximise the log-likelihood
 - ▶ $\ell(\nu, m, s) = \sum_{t=1}^n \ln f((r_t - m)/s; \nu) - n \ln s$
 - ▶ no closed form: a numerical optimiser (`scipy.stats.t.fit` in the notebook)
- **AIC** (Akaike information criterion) [Akaike \(1974\)](#): $AIC = 2k - 2\ell$
 - ▶ k = number of parameters (2 for the Normal distribution, 3 for the t); the lower AIC wins
 - ▶ $\Delta AIC = AIC_{\text{Normal}} - AIC_t > 0$ favours the t
- Early evidence for stock prices: [Praetz \(1972\)](#), [Blattberg & Gonedes \(1974\)](#)

Fitted Student- t : Degrees of Freedom, 2010–2026

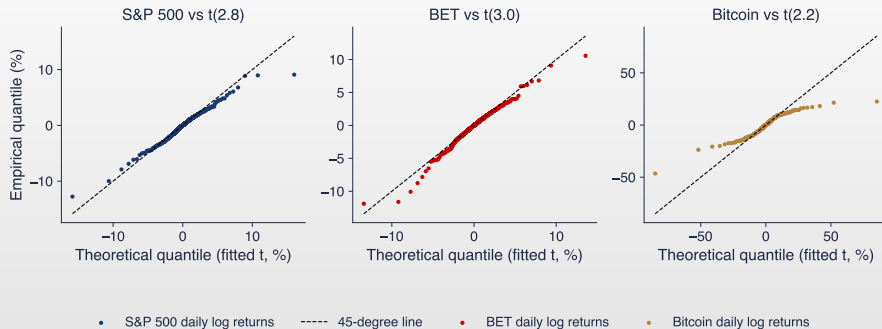
Series	$\hat{\nu}$	JB	ΔAIC
S&P 500	2.78	32 113	1213
DAX	3.31	9 909	739
BET	2.96	56 762	1454
Bitcoin	2.23	26 482	1335
OMV Petrom (SNP)	3.79	12 832	679
BRD	4.67	8 370	478
Nuclearelectrica (SNN)	3.08	3 174	541
Romgaz (SNG)	3.40	3 415	482

- All $\hat{\nu}$ between 2.23 (Bitcoin) and 4.67 (BRD)
- $\hat{\nu} < 4$ for 7 of the 8 series: the fitted model has infinite kurtosis
- ΔAIC is at least 478: the t beats the Normal distribution everywhere

 SFM_ch2_student_t_fit

S&P 500: Normal vs Student- t Fit

- Linear scale: the t ($\hat{\nu} = 2.78$) follows the tall centre that the Normal distribution misses
- Log scale: the Normal density collapses beyond $\pm 4\%$; the t follows the observed tails

QQ Plots against the Fitted Student- t 

- The centre and the moderate tails now lie on the line
- Extremes: the fitted t expects lower minima than observed: S&P 500 -15.8% vs -12.8% ; Bitcoin -85.0% vs -46.5%
- With $\hat{\nu}$ close to 2 the model overshoots the most extreme quantiles

What the Student- t Still Misses

- **Symmetry:** the t cannot reproduce the negative skewness
 - ▶ skewed t distributions exist; they are beyond this chapter
- **Independence:** an i.i.d. t model has no volatility clustering
 - ▶ GARCH models with t errors combine both [Bollerslev \(1987\)](#) ([Chapter 9](#))
- **Single tail index:** one ν for both tails and for the centre
 - ▶ extreme value theory models each tail separately ([Chapter 5](#))
- Still the simplest heavy-tailed model, and a large improvement on the Normal distribution

Recap: Student- t

- One extra parameter ν controls the tails; power-law tails, the Normal distribution as $\nu \rightarrow \infty$
- Fitted $\hat{\nu}$ between 2 and 5 for daily returns; AIC strongly prefers the t
- It misses skewness and volatility clustering

What Is a Stylised Fact?

- ▣ A **stylised fact**: a statistical property shared by many assets, markets and periods [Cont \(2001\)](#)
- ▣ Qualitative, not exact: “heavy tails”, not “ $\nu = 3.2$ ”
- ▣ Rama Cont lists eleven; we check six on our data
- ▣ A good model of returns must reproduce them; the benchmark i.i.d. Normal model fails most of them
- ▣ Earlier evidence: [Mandelbrot \(1963\)](#), [Fama \(1965\)](#)



Rama Cont

Photo: [Renate Schmid \(2012\)](#); CC BY-SA 2.0 DE;
[Wikimedia Commons](#)

Six Stylised Facts

- ▣ 1. **Heavy tails:** large moves are far more frequent than under the Normal distribution
- ▣ 2. **Aggregational Gaussianity:** returns over longer horizons are closer to the Normal distribution
- ▣ 3. **Absence of linear autocorrelation:** past returns barely predict future returns
- ▣ 4. **Volatility clustering:** large moves follow large moves, of either sign
- ▣ 5. **Leverage effect:** falls raise future volatility more than rises
- ▣ 6. **Gain/loss asymmetry:** large losses are larger and more frequent than large gains

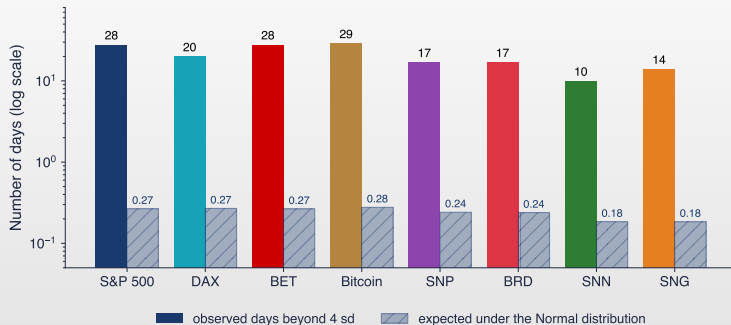
Fact 1: Heavy Tails, Counted

Series	n	$> 3\sigma$	$> 4\sigma$	$> 5\sigma$	$> 6\sigma$	p (Poisson, 4σ)
S&P 500	4 201	62	28	12	8	2×10^{-46}
DAX	4 244	56	20	5	4	1×10^{-30}
BET	4 186	70	28	16	11	2×10^{-46}
Bitcoin	4 384	80	29	10	4	6×10^{-48}
Normal distribution, $n = 4\,201$		11.34	0.27	2.4×10^{-3}	8.3×10^{-6}	

- $> k\sigma$: number of days with $|r_t - \bar{r}| > k\sigma$; last row: number expected under the Normal distribution
- p : probability of at least that many 4-sigma days if the Normal count is Poisson with mean $n \times 6.3 \times 10^{-5}$
- S&P 500: 105 times more 4-sigma days than the Normal distribution allows

 SFM_ch2_sigma_days

Fact 1: 4-Sigma Days in Every Market

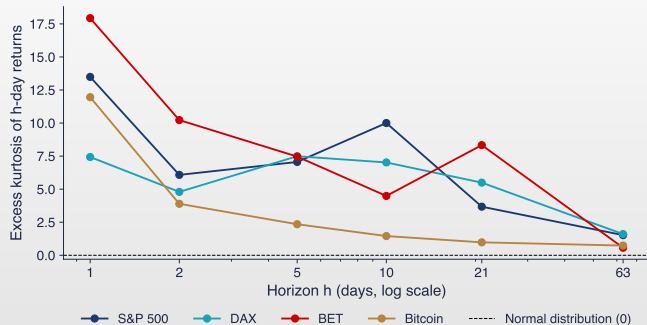


- Observed days beyond 4 standard deviations (log scale) against the Normal expectation of 0.18–0.28 days
- Indices and Bitcoin: one 4-sigma day every 150–212 days; BVB stocks: 10–17 days in total
- Under a fitted Student- t , the expected numbers are close to the observed ones: S&P 500 34.9, BET 28.1 (Seminar 2, B4)

Fact 2: Aggregational Gaussianity

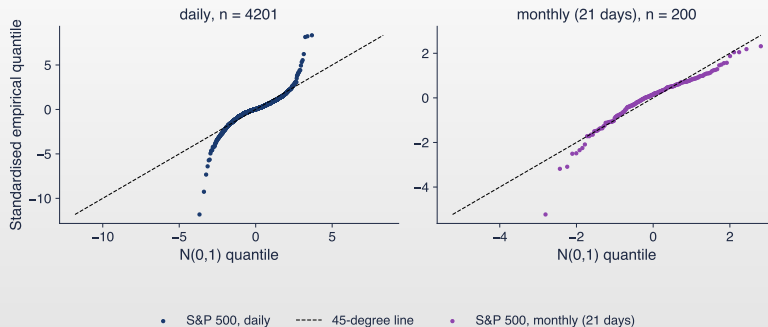
- h -day log return: $r_t(h) = r_t + r_{t-1} + \dots + r_{t-h+1}$, taken on non-overlapping blocks
 - ▶ if the CLT applies, the distribution of $r_t(h)$ approaches the Normal distribution as h grows
- The stylised fact: the shape changes with the horizon; it is not the same at all time scales [Cont \(2001\)](#)
- Price to pay: fewer observations
 - ▶ S&P 500, 17 years: 4 201 daily, 200 monthly and 66 quarterly returns
 - ▶ estimates of kurtosis at long horizons are very noisy

Fact 2: Kurtosis Falls with the Horizon



- Excess kurtosis, daily $\rightarrow$ monthly $\rightarrow$ quarterly: S&P 500 13.5 $\rightarrow$ 3.7 $\rightarrow$ 1.5; BET 17.9 $\rightarrow$ 8.3 $\rightarrow$ 0.6; Bitcoin 12.0 $\rightarrow$ 1.0 $\rightarrow$ 0.7
- The fall is slow and not monotone: one crash month (March 2020) dominates a sample of 200 months
- Quarterly returns: JB no longer rejects normality for the BET ($p = 0.07$) and Bitcoin ($p = 0.22$)

Fact 2: S&P 500, Daily vs Monthly QQ Plots

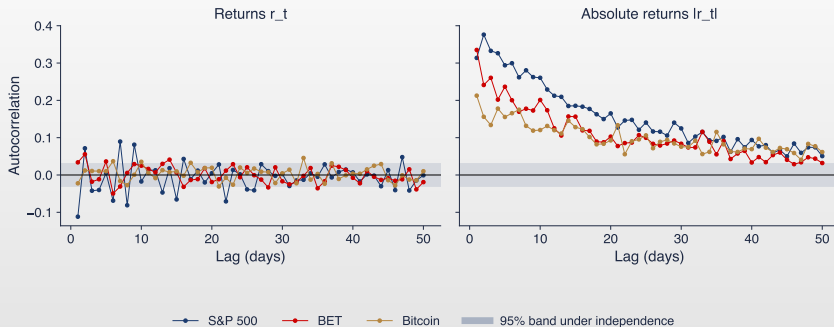


- Monthly returns lie much closer to the line than daily returns
- But monthly returns are not Normal: skewness -1.18 , excess kurtosis 3.7 , $n = 200$; the lowest point is March 2020
- Why so slow? Volatility clustering makes daily returns dependent, which slows the CLT

Fact 3: Absence of Linear Autocorrelation

- **ACF** (autocorrelation function): $\rho(h) = \text{Corr}(r_t, r_{t-h})$, $h = 1, 2, \dots$
 - ▶ sample version $\hat{\rho}(h) = \sum_t (r_t - \bar{r})(r_{t-h} - \bar{r}) / \sum_t (r_t - \bar{r})^2$
 - ▶ for i.i.d. data, $\hat{\rho}(h) \approx N(0, 1/n)$: 95% band $\pm 1.96/\sqrt{n}$
- **LB** (Ljung–Box) test [Ljung & Box \(1978\)](#) of $H_0: \rho(1) = \dots = \rho(m) = 0$
 - ▶ $Q(m) = n(n+2) \sum_{h=1}^m \hat{\rho}(h)^2 / (n-h) \sim \chi^2(m)$; for $m = 10$, reject if $Q > 18.3$
- Why absent? If returns were predictable, traders would exploit it until the predictability disappears (efficient markets, [Chapter 7](#))

Facts 3 and 4: ACF of Returns and of Absolute Returns



- Left: returns; most autocorrelations are inside or close to the band; S&P 500 $\hat{\rho}(1) = -0.11$, BET 0.03, Bitcoin -0.02
- Right: absolute returns; all positive, slowly decaying: S&P 500 0.31 at lag 1, 0.26 at lag 10, 0.05 at lag 50

Fact 3: Small but Not Always Zero

- Ljung–Box $Q(10)$ for returns: S&P 500 199, BET 46, Bitcoin 20; critical value 18.3
 - ▶ formally, some autocorrelation is significant
- Two reasons not to over-read this
 - ▶ the values are economically small: $|\hat{\rho}(h)| \leq 0.11$, so at most 1.2% of the variance is explained
 - ▶ the band $\pm 1.96/\sqrt{n}$ assumes i.i.d. returns; with volatility clustering it is too narrow, so “significant” is found too often
- The S&P 500 value at lag 1 comes mostly from March 2020, when large falls and rebounds alternated: without 20 February–30 April 2020, $\hat{\rho}(1) = -0.04$
- Robust tests of predictability (variance ratios): [Chapter 7](#)

Fact 4: Volatility Clustering

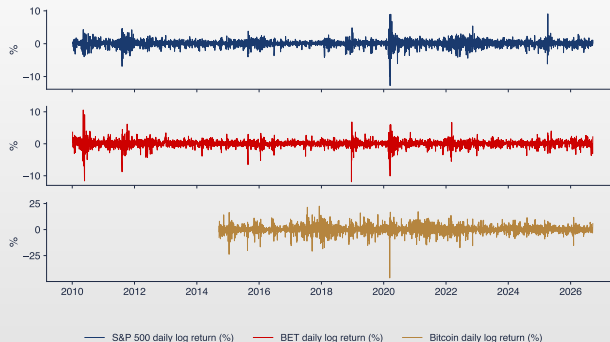
- “Large changes tend to be followed by large changes, of either sign, and small changes tend to be followed by small changes” [Mandelbrot \(1963\)](#)
- The sign of tomorrow’s return is unpredictable; its size is not
- Evidence: ACF of $|r_t|$ positive for 50 lags
 - ▶ Ljung–Box $Q(10)$ of $|r_t|$: S&P 500 3859, BET 2120, Bitcoin 1086
- So returns are uncorrelated but **not independent**
- Models: ARCH and GARCH ([Chapter 9](#))



Benoit Mandelbrot (1924–2010)

Photo: [Rama \(2007\)](#); CC BY-SA 2.0 fr; [Wikimedia Commons](#)

Fact 4: Calm and Turbulent Periods

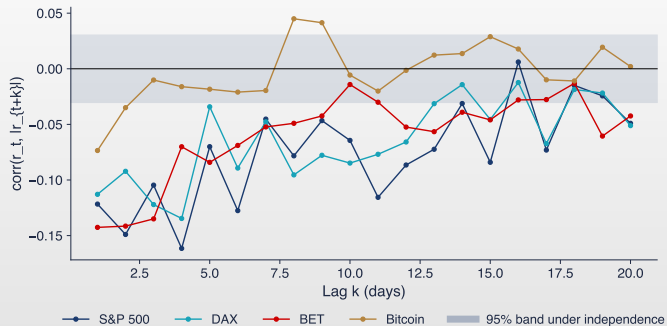


- Turbulent periods: 2010–2011 (euro area debt crisis), December 2018 for the BET, March 2020 everywhere, April 2025
- Calm years in between: the volatility of a single day depends on the period it falls in

Fact 5: The Leverage Effect

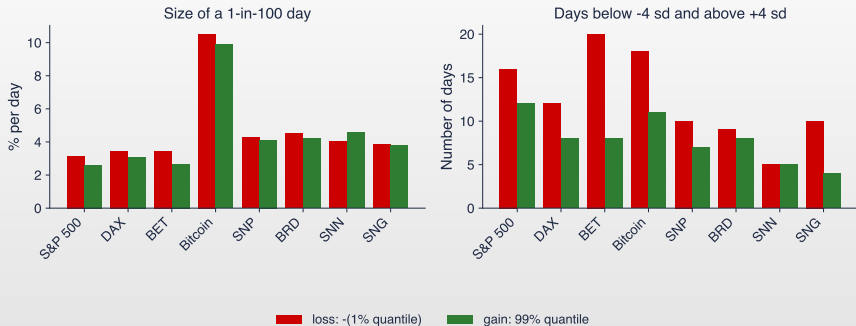
- ▣ Measure: $L(k) = \text{Corr}(r_t, |r_{t+k}|)$, $k = 1, 2, \dots$
 - ▶ $L(k) < 0$: a fall today is followed by larger moves than a rise of the same size
- ▣ Explanation by financial leverage [Christie \(1982\)](#)
 - ▶ when the share price falls, the debt-to-equity ratio rises, so equity becomes riskier
- ▣ A second explanation, volatility feedback: higher expected volatility raises required returns and lowers prices
- ▣ Models with an asymmetric response (EGARCH, GJR-GARCH): [Chapter 9](#)

Fact 5: Leverage Correlations



- Stock indices: $L(k) < 0$ for most of the first 20 lags; mean of $L(1..5)$: S&P 500 -0.12 , DAX -0.10 , BET -0.11
- Bitcoin: $L(1) = -0.07$, then inside the band: no firm has debt behind Bitcoin, so the leverage mechanism is absent

Fact 6: Gain/Loss Asymmetry



- Left: a 1-in-100 loss is larger than a 1-in-100 gain for the indices: BET 3.45% vs 2.66%, S&P 500 3.15% vs 2.61%
- Right: days below -4σ vs above $+4\sigma$: BET 20 vs 8, S&P 500 16 vs 12, Bitcoin 18 vs 11
- Exception: SNN, with positive skewness and a larger right quantile

Six Facts, Eight Series

Series	Exc. kurt.	Exc. kurt., 21 days	$\hat{\rho}_r(1)$	$\hat{\rho}_{ r }(10)$	$\bar{L}(1..5)$	$ q_{0.01} /q_{0.99}$
S&P 500	13.5	3.7	-0.11	0.26	-0.12	1.21
DAX	7.4	5.5	0.01	0.16	-0.10	1.12
BET	17.9	8.3	0.03	0.20	-0.11	1.30
Bitcoin	12.0	1.0	-0.02	0.12	-0.03	1.07
OMV Petrom (SNP)	8.9	3.3	-0.02	0.13	-0.03	1.04
BRD	7.3	0.4	0.03	0.11	-0.05	1.07
Nuclearelectrica (SNN)	5.1	2.3	0.09	0.08	-0.03	0.88
Romgaz (SNG)	5.3	1.4	0.07	0.08	-0.04	1.02

- Facts 1, 3 and 4 hold for every series; fact 2 holds with noise; facts 5 and 6 hold for the indices and most BVB stocks, weakly or not for Bitcoin and SNN
- A comparison of the return distributions of cryptocurrencies and traditional assets: [Pele et al. \(2023\)](#)

 SFM_ch2_cont_stylised_facts

What the Facts Mean for Models

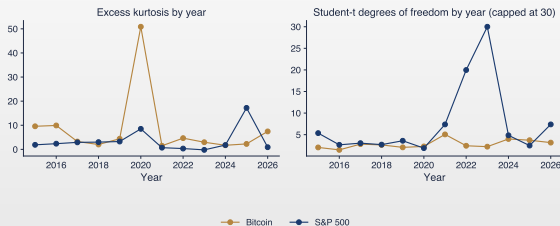
Model	1	2	3	4	5	6
i.i.d. Normal	–	trivially	✓	–	–	–
i.i.d. Student- t	✓	✓	✓	–	–	–
GARCH with t errors (Ch. 9)	✓	✓	✓	✓	–	–
EGARCH, GJR-GARCH (Ch. 9)	✓	✓	✓	✓	✓	(✓)

- Columns: the six facts in the order of this section; (✓): only for returns over several days, unless the errors are skewed
- Risk measures (VaR 1%, ES 2.5%, [Chapter 10](#)) inherit the quality of the model: a Normal VaR is too low in crises
- A related summary of tail risk, information entropy: [Pele, Lazar & Dufour \(2017\)](#)

Recap: Stylised Facts

- ▣ Heavy tails: 4-sigma days every few months instead of once in decades
- ▣ Returns are nearly uncorrelated, but their size is strongly autocorrelated
- ▣ Longer horizons are closer to the Normal distribution, slowly
- ▣ Stock indices: falls raise volatility, and losses are larger than gains

An Open Question: Are Bitcoin's Tails Getting Thinner?



- As a market matures (more traders, futures, ETFs), its tails might thin out
- Bitcoin, 2015–2026: yearly $\hat{\nu}$ between 1.5 and 5.1; rank correlation with the year 0.63 ($p = 0.03$)
- Why it is open: 12 noisy yearly estimates, one year (2020, excess kurtosis 51) dominated by one day
- AI tools can speed up such a study; they do not replace checking it [Wang et al. \(2023\)](#)

 [SFM_ch2_tails_by_year](#)

How AI Could Help

- ▣ **Literature:** find studies of tail indices of crypto assets and summarise their methods
- ▣ **Code:** draft rolling-window estimates of ν and of the excess kurtosis
- ▣ **Robustness:** propose other windows, other tail measures ([Chapter 5](#)), other crypto assets
- ▣ Example prompt
 - ▶ `Write a Python function that fits a Student-t distribution to daily Bitcoin log returns on rolling 365-day windows and returns the degrees of freedom with a bootstrap 95% interval.`

What to Check

- ▣ Data: the same price column and calendar for every year; errors among the largest moves
- ▣ Estimation: did the optimiser converge? $\hat{\nu}$ near 2 or above 30 needs a second look
- ▣ Inference: one $\hat{\nu}$ per year is noisy; report intervals, not only point estimates
- ▣ Volatility clustering: thinner yearly tails may only mean calmer years ([Chapter 9](#))
- ▣ References: every cited paper must exist; check the DOI

Project Seed

- ▣ **Question:** do the tails of crypto assets thin out as their markets mature?
 - ▶ data: Bitcoin, Ethereum and Solana from the course data; S&P 500 as a benchmark
- ▣ Steps
 - ▶ estimate $\hat{\nu}$ and the excess kurtosis on rolling one-year windows, with bootstrap intervals
 - ▶ test for a trend; repeat on returns divided by a rolling volatility
 - ▶ compare the count of 4-sigma days per year with the S&P 500
- ▣ Deliverable: one table, one chart, and a paragraph on what the data can and cannot show
- ▣ Declare any AI use, and list the errors of the AI that you corrected

Key Takeaways

- ▣ The Normal distribution: two parameters, thin tails, closed under addition
- ▣ Normal log returns give lognormal prices; mean above median by the volatility drag
- ▣ The CLT needs independence, identical distribution and finite variance
- ▣ Daily returns: mostly negative skewness, excess kurtosis 5–18, JB rejects, QQ plots bend at both ends
- ▣ The Student- t with $\hat{\nu}$ between 2 and 5 fits the tails much better, but is symmetric and i.i.d.
- ▣ Six stylised facts: heavy tails, aggregational Gaussianity, no linear autocorrelation, volatility clustering, leverage, gain/loss asymmetry

Key Formulas

Quantity	Formula
Normal PDF	$f(x) = (\sigma\sqrt{2\pi})^{-1} \exp\left(- (x - \mu)^2 / (2\sigma^2)\right)$
Lognormal moments	median e^m , mean $e^{m+s^2/2}$
CLT	$\sqrt{n}(\bar{X}_n - \mu)/\sigma \xrightarrow{d} N(0, 1)$
Skewness, excess kurtosis	$S = m_3/m_2^{3/2}$, $K = m_4/m_2^2 - 3$
Jarque–Bera	$JB = \frac{n}{6}(S^2 + K^2/4) \sim \chi^2(2)$
Student- $t(\nu)$	variance $\nu/(\nu - 2)$, excess kurtosis $6/(\nu - 4)$
AIC	$2k - 2\ell$
ACF, Ljung–Box	$\hat{\rho}(h)$, $Q(m) = n(n+2) \sum_{h=1}^m \hat{\rho}(h)^2 / (n-h)$
Leverage	$L(k) = \text{Corr}(r_t, r_{t+k})$

Check Yourself

- **Question:** daily returns are $N(0.05\%, 1\%^2)$; how many standard deviations is a -3% day?
▶ **Answer:** $z = (-3 - 0.05)/1 = -3.05$
- **Question:** $n = 2500$, $S = -0.5$, $K = 6$; what is JB?
▶ **Answer:** $\frac{2500}{6}(0.25 + 9) = 3854$; normality is rejected
- **Question:** an annual log return is $N(5\%, 20\%^2)$; what are the median and the mean of 100 invested after one year?
▶ **Answer:** median $100e^{0.05} = 105.1$, mean $100e^{0.05+0.02} = 107.3$
- **Next:** [Chapter 3](#), α -stable distributions

References (1/2)

- Akaike, H. (1974). A new look at the statistical model identification. *IEEE Transactions on Automatic Control*, 19(6), 716–723.
- Bachelier, L. (1900). Théorie de la spéculation. *Annales scientifiques de l'École Normale Supérieure*, 17, 21–86.
- Blattberg, R.C., Gonedes, N.J. (1974). A comparison of the stable and Student distributions as statistical models for stock prices. *Journal of Business*, 47(2), 244–280.
- Bollerslev, T. (1987). A conditionally heteroskedastic time series model for speculative prices and rates of return. *Review of Economics and Statistics*, 69(3), 542–547.
- Borak, S., Härdle, W.K., López-Cabrera, B. (2013). *Statistics of Financial Markets: Exercises and Solutions*, 2nd ed. Springer.
- Christie, A.A. (1982). The stochastic behavior of common stock variances: value, leverage and interest rate effects. *Journal of Financial Economics*, 10(4), 407–432.
- Cont, R. (2001). Empirical properties of asset returns: stylized facts and statistical issues. *Quantitative Finance*, 1(2), 223–236.
- Fama, E.F. (1965). The behavior of stock-market prices. *Journal of Business*, 38(1), 34–105.
- Franke, J., Härdle, W.K., Hafner, C.M. (2019). *Statistics of Financial Markets: An Introduction*, 5th ed. Springer.
- Jarque, C.M., Bera, A.K. (1980). Efficient tests for normality, homoscedasticity and serial independence of regression residuals. *Economics Letters*, 6(3), 255–259.
- Jarque, C.M., Bera, A.K. (1987). A test for normality of observations and regression residuals. *International Statistical Review*, 55(2), 163–172.

References (2/2)

- Ljung, G.M., Box, G.E.P. (1978). On a measure of lack of fit in time series models. *Biometrika*, 65(2), 297–303.
- Mandelbrot, B. (1963). The variation of certain speculative prices. *Journal of Business*, 36(4), 394–419.
- Osborne, M.F.M. (1959). Brownian motion in the stock market. *Operations Research*, 7(2), 145–173.
- Pele, D.T., Lazar, E., Dufour, A. (2017). Information entropy and measures of market risk. *Entropy*, 19(5), 226.
- Pele, D.T., Wesselhöfft, N., Härdle, W.K., Kolossiatis, M., Yatracos, Y.G. (2023). Are cryptos becoming alternative assets? *European Journal of Finance*, 29(10), 1064–1105.
- Praetz, P.D. (1972). The distribution of share price changes. *Journal of Business*, 45(1), 49–55.
- Student (1908). The probable error of a mean. *Biometrika*, 6(1), 1–25.
- Tsay, R.S. (2010). *Analysis of Financial Time Series*, 3rd ed. Wiley.
- Wang, H., Fu, T., Du, Y., Gao, W., et al. (2023). Scientific discovery in the age of artificial intelligence. *Nature*, 620, 47–60.
- Wilk, M.B., Gnanadesikan, R. (1968). Probability plotting methods for the analysis of data. *Biometrika*, 55(1), 1–17.